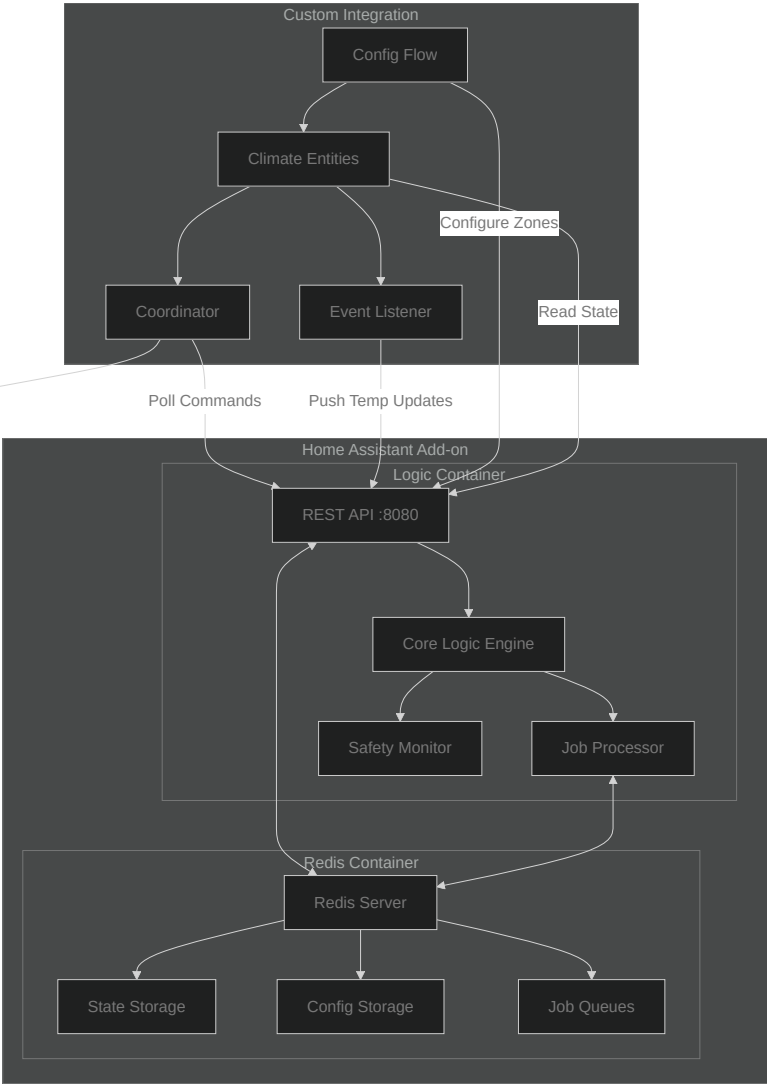
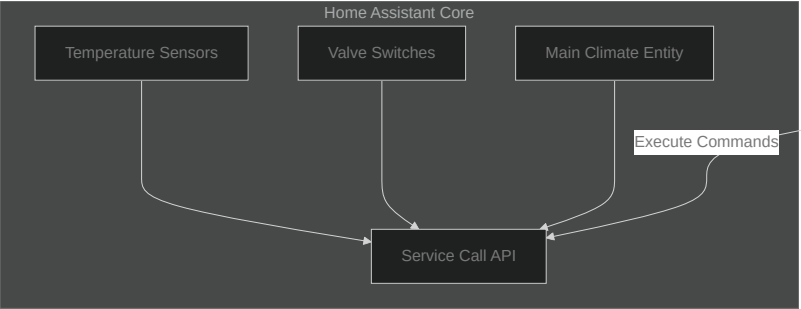
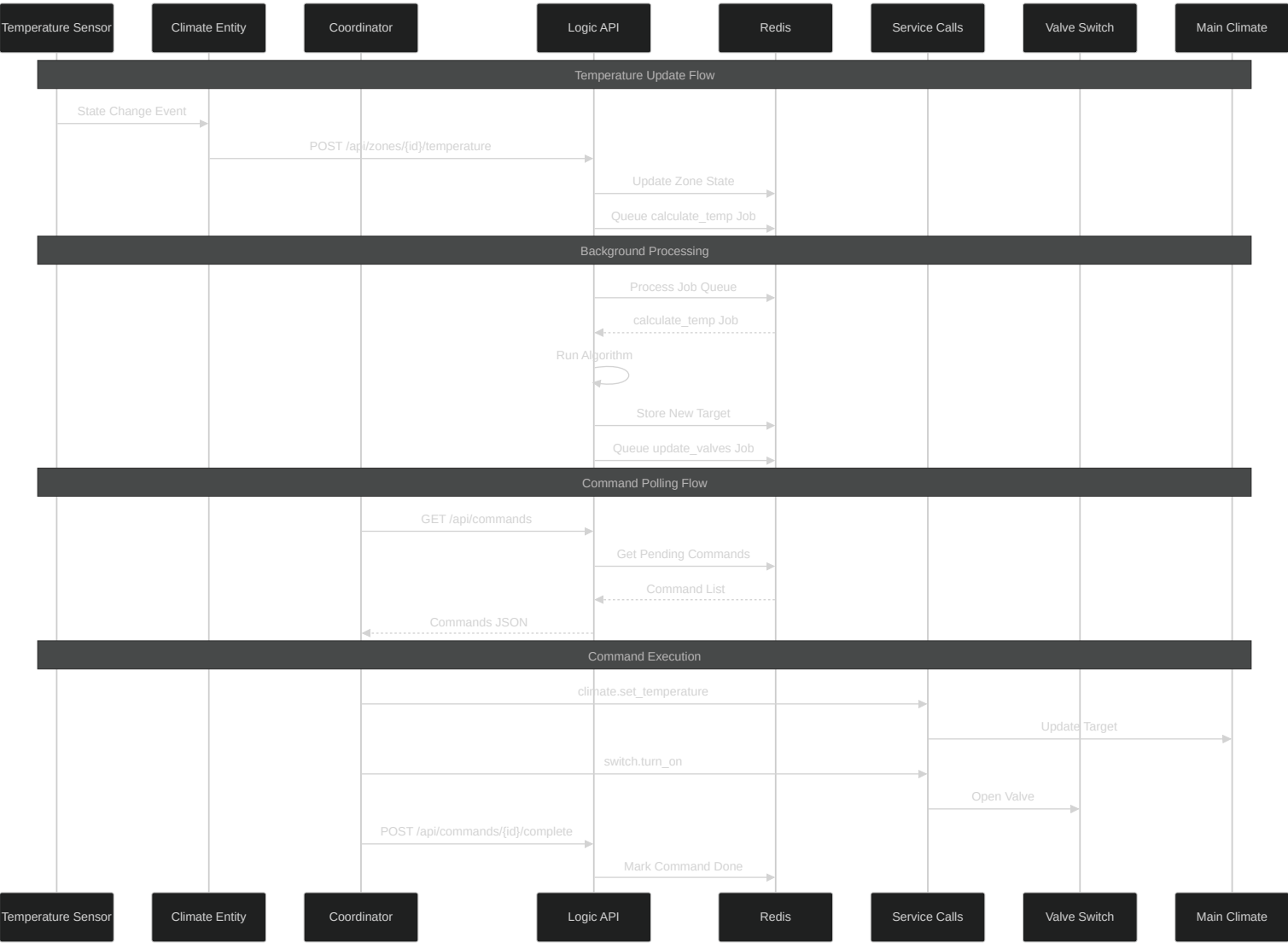
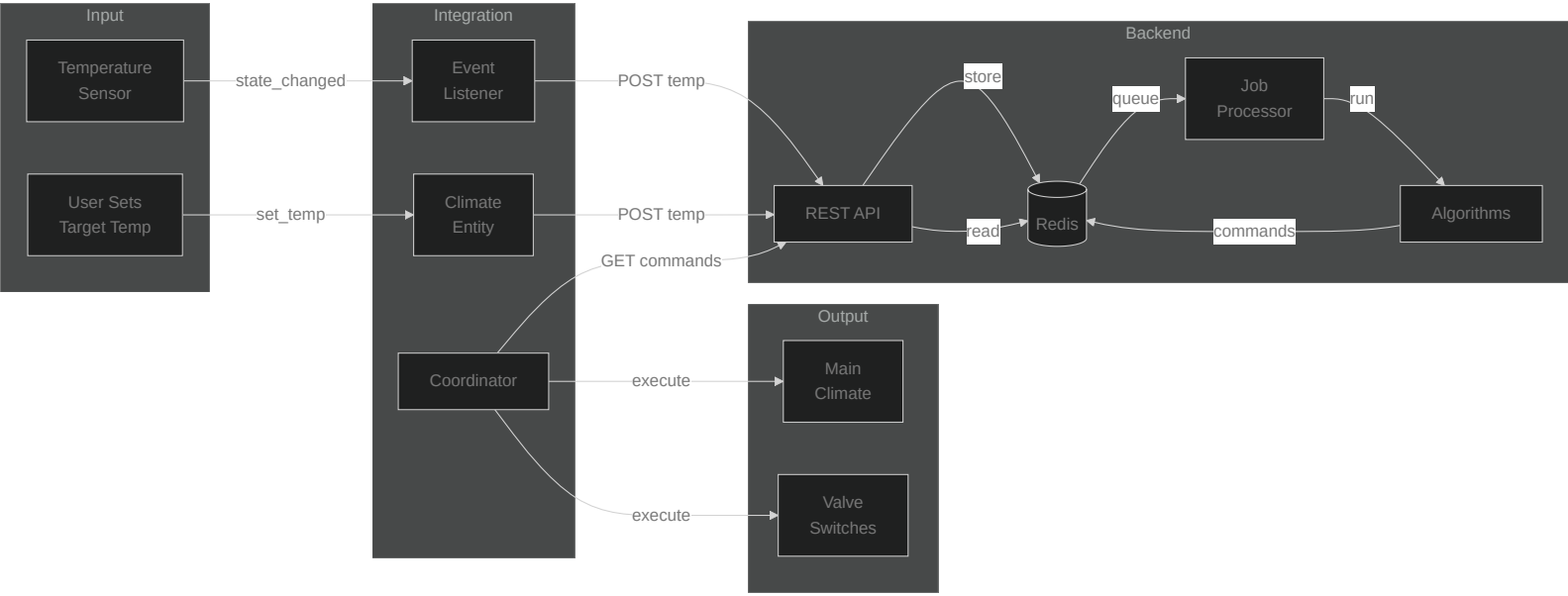
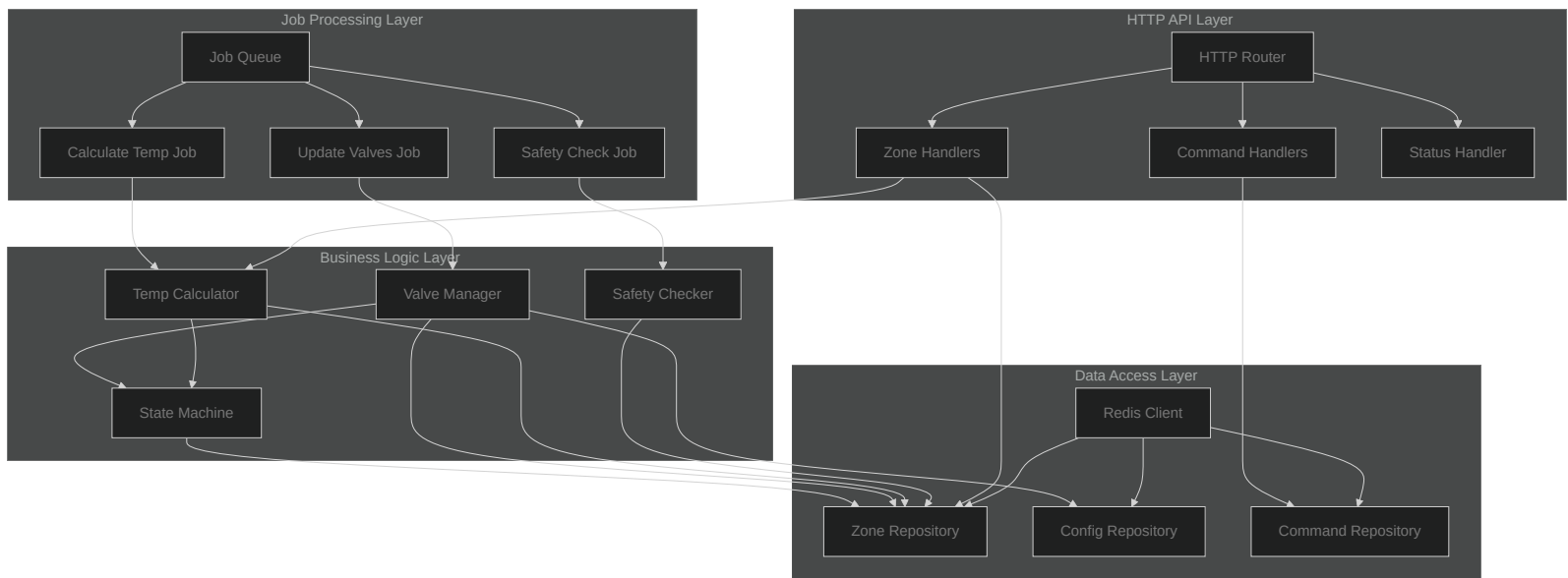
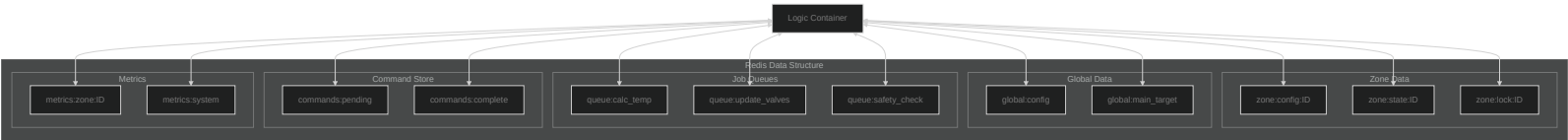


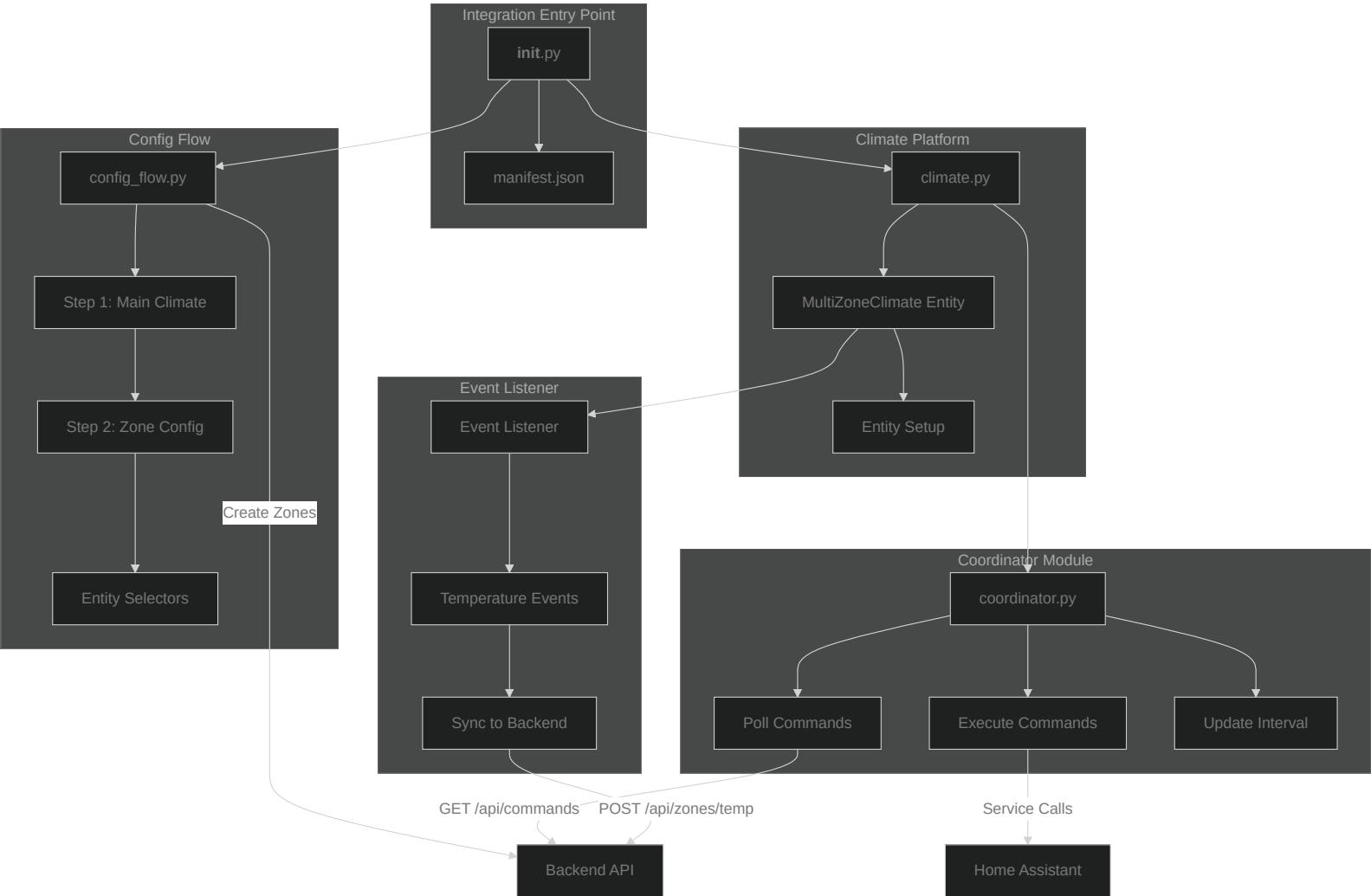




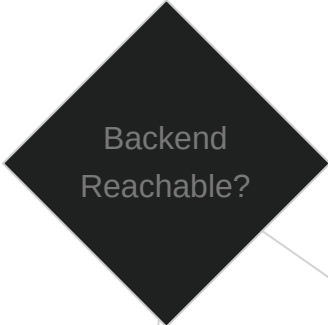








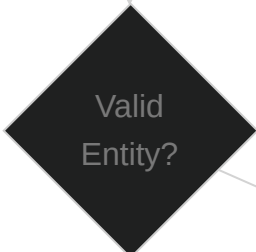
User Adds Integration



Yes

Step 1: Select Main Climate

Show Climate Entity
Selector



No

Show Error

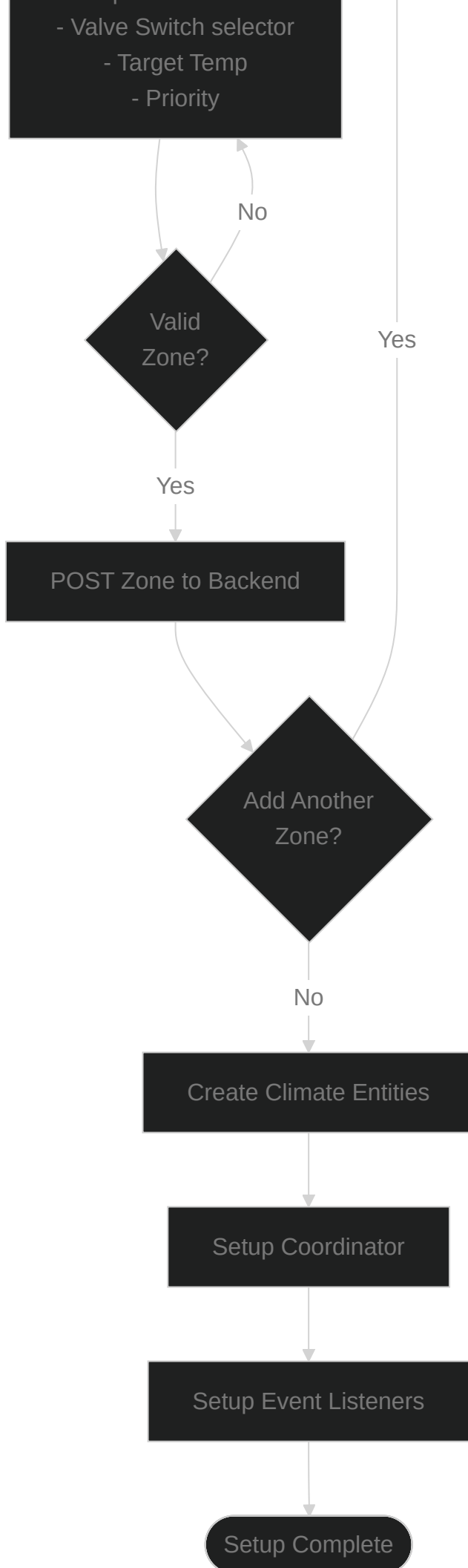
Yes

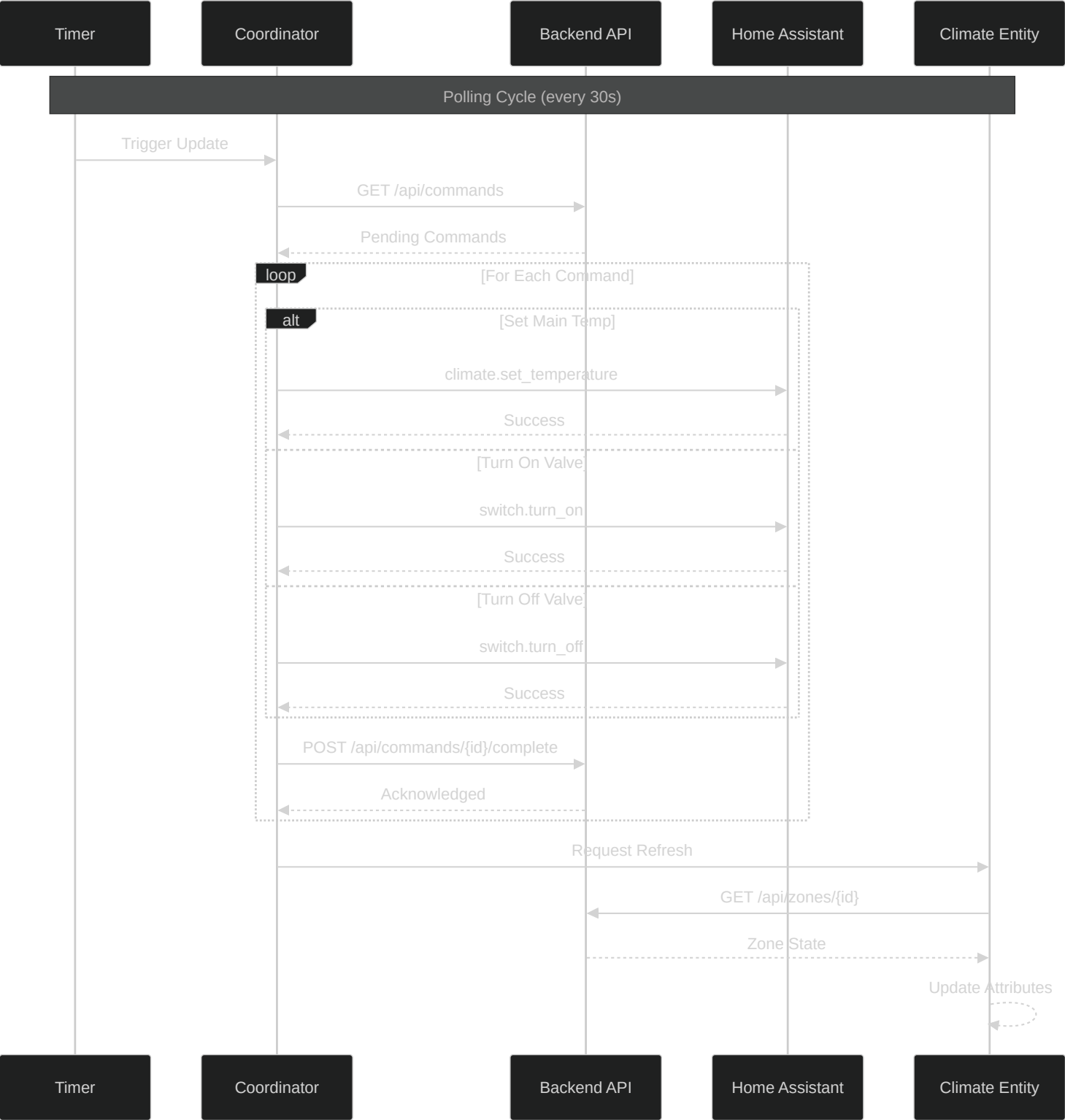
Store Main Climate ID

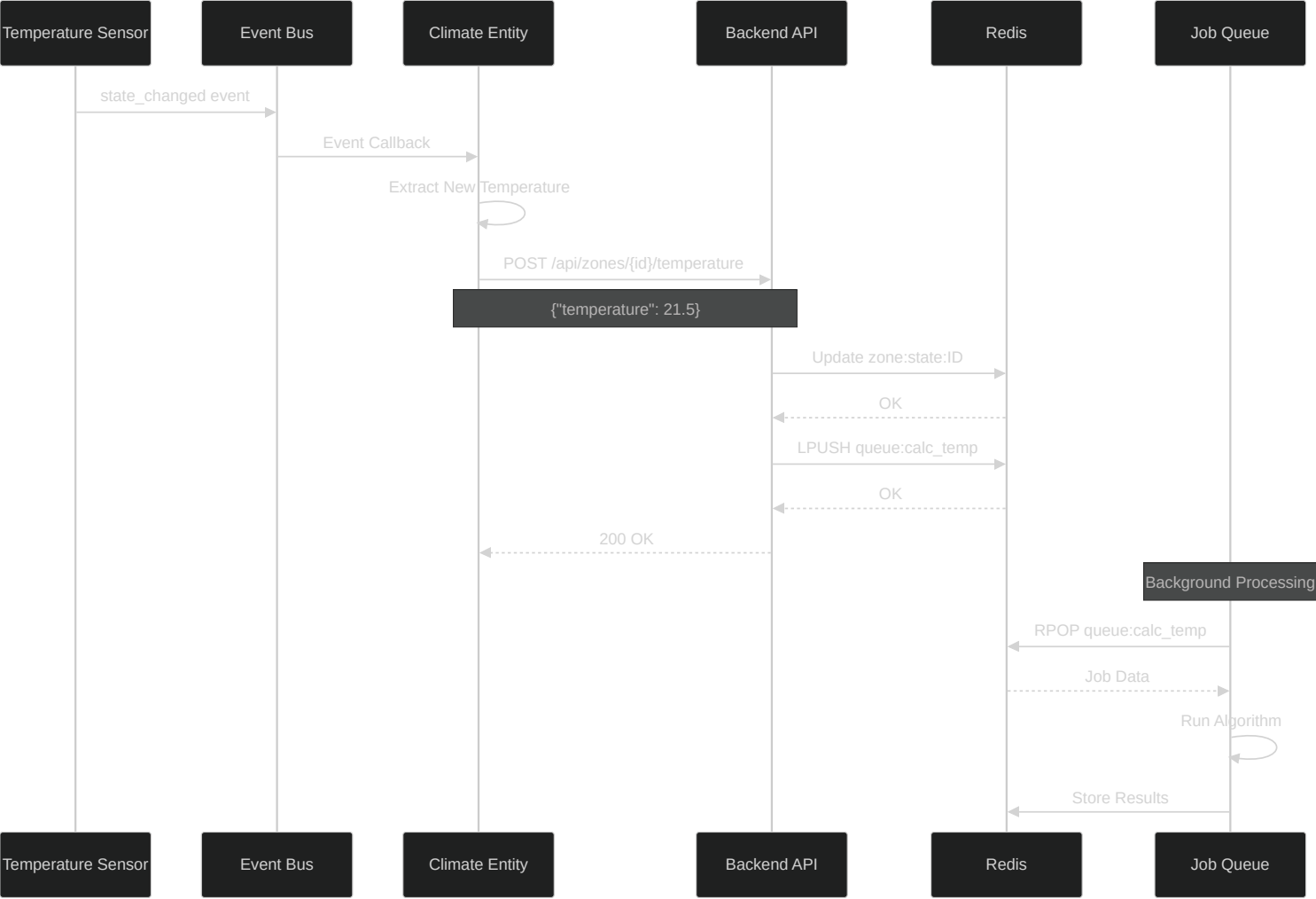
Step 2: Configure Zone

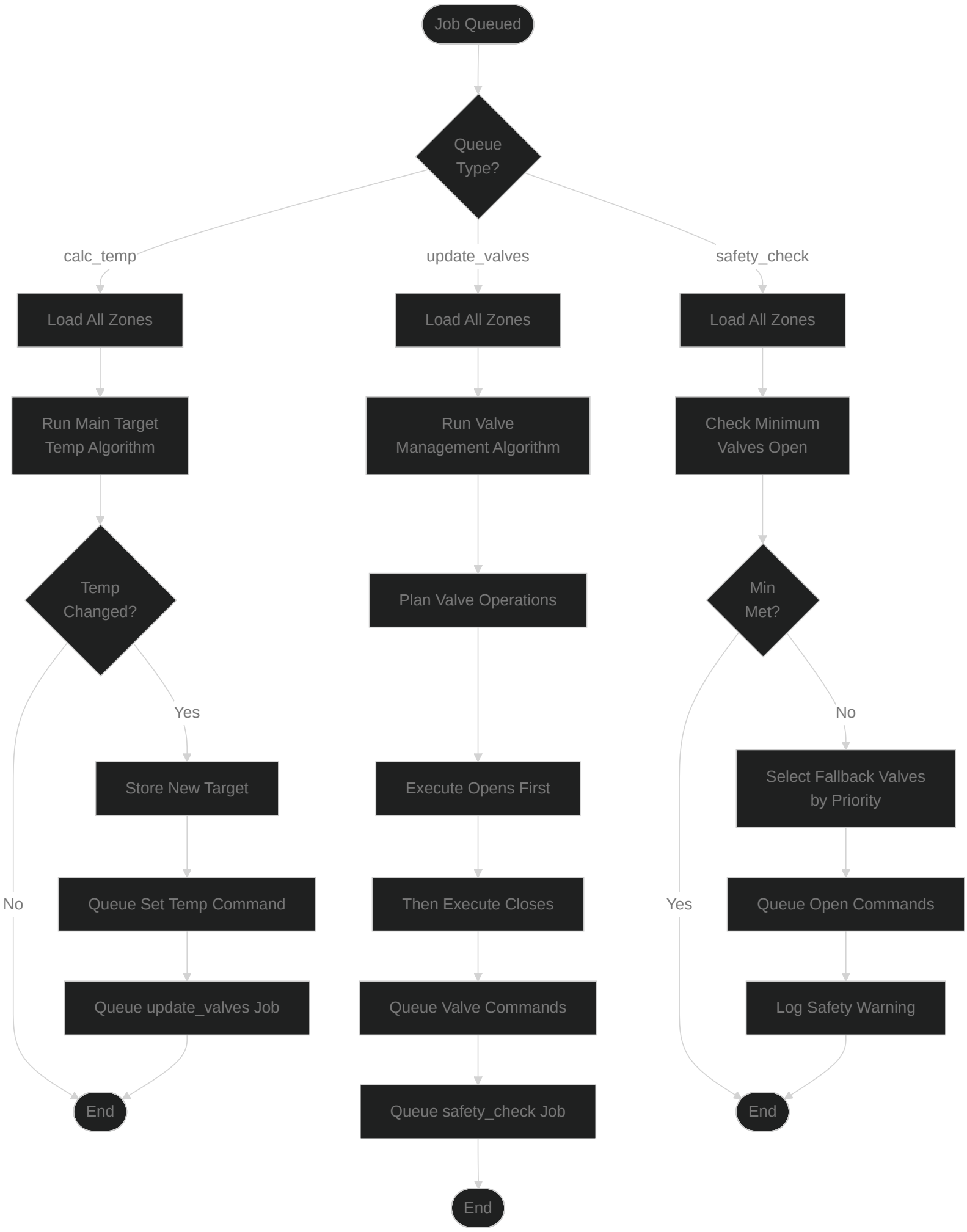
Show Zone Form:
- Name
- Temp Sensor selector

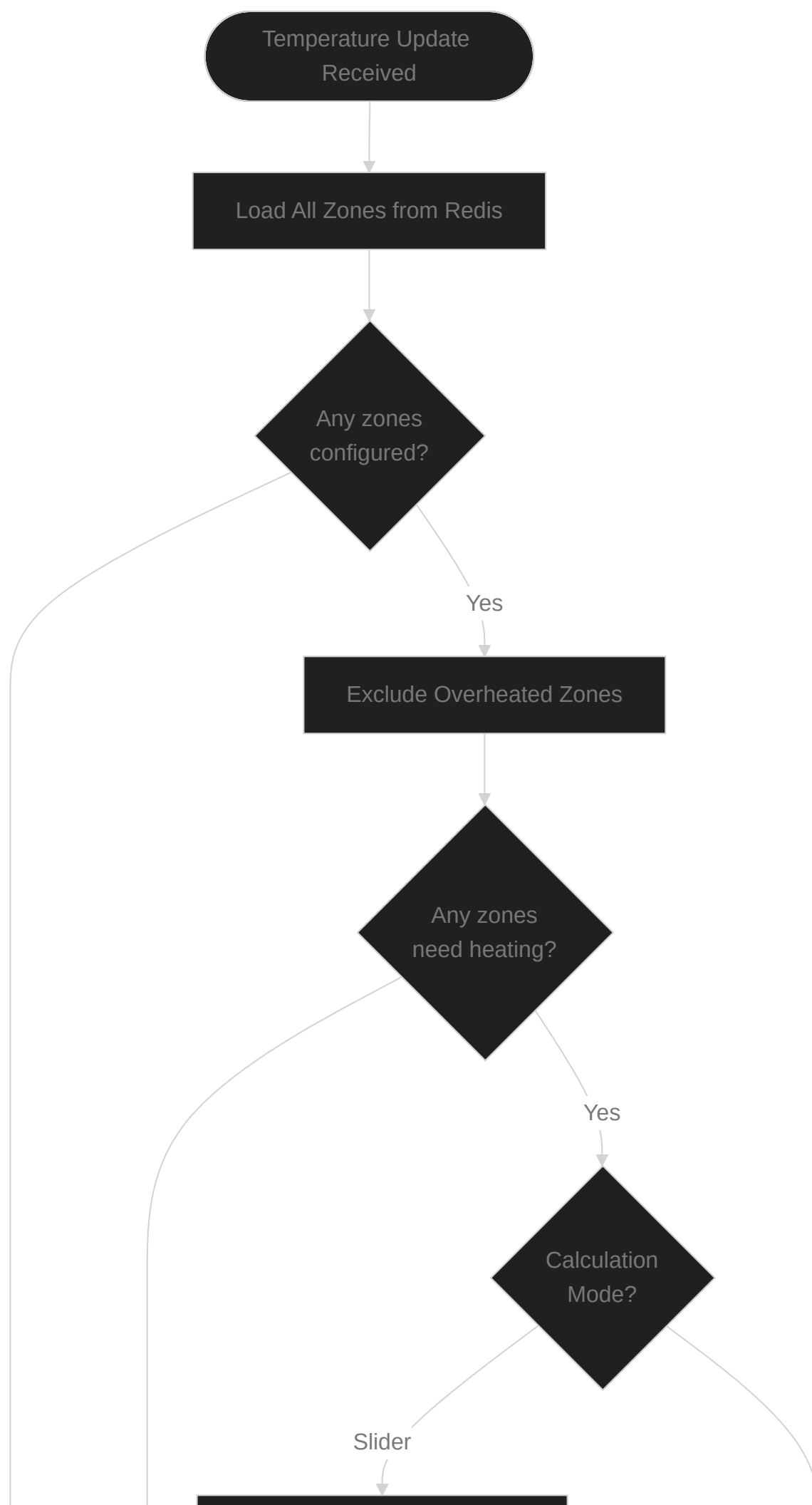
No

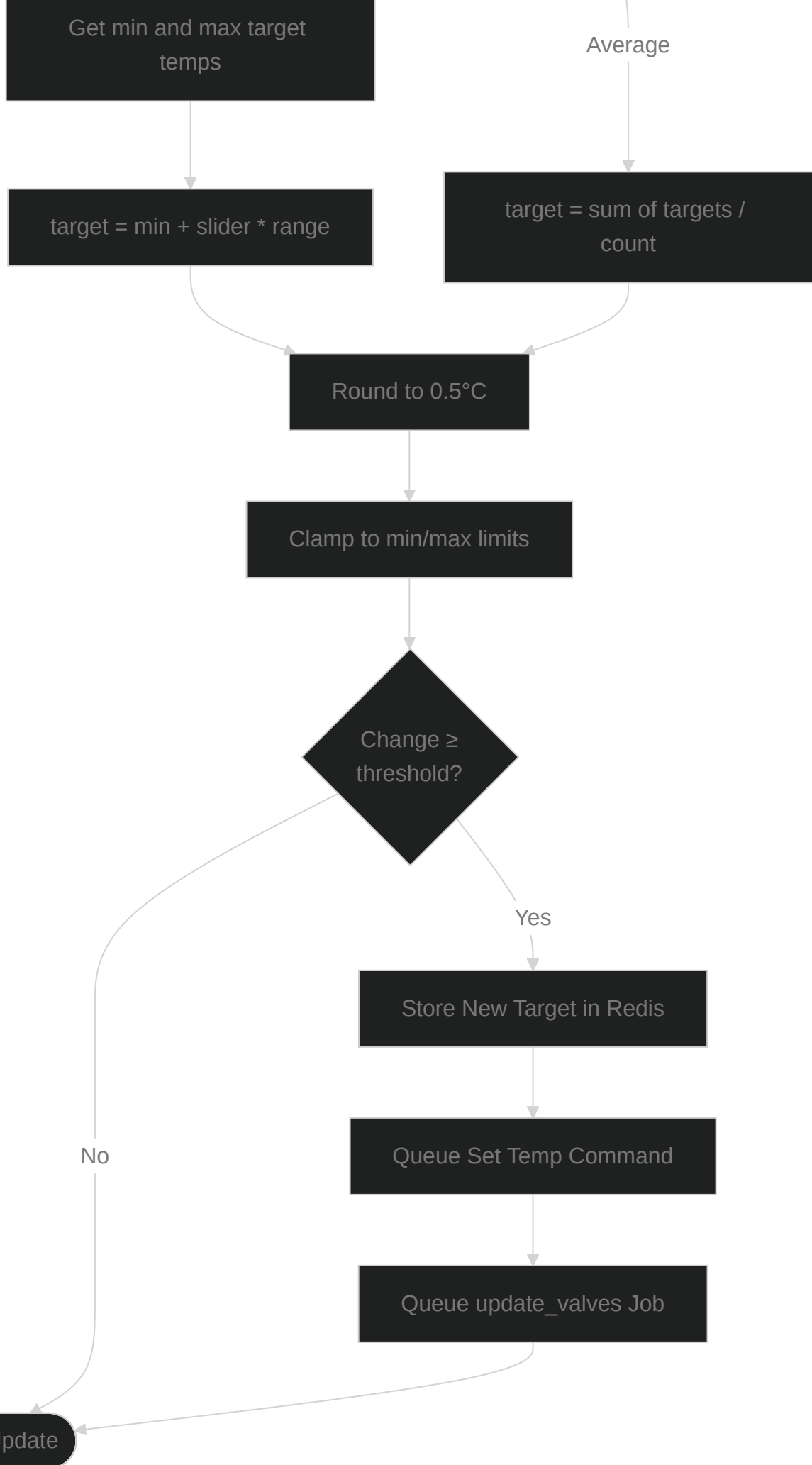












No

No

No

Average

target = sum of targets /
count

target = min + slider * range

Round to 0.5°C

Clamp to min/max limits

Change ≥
threshold?

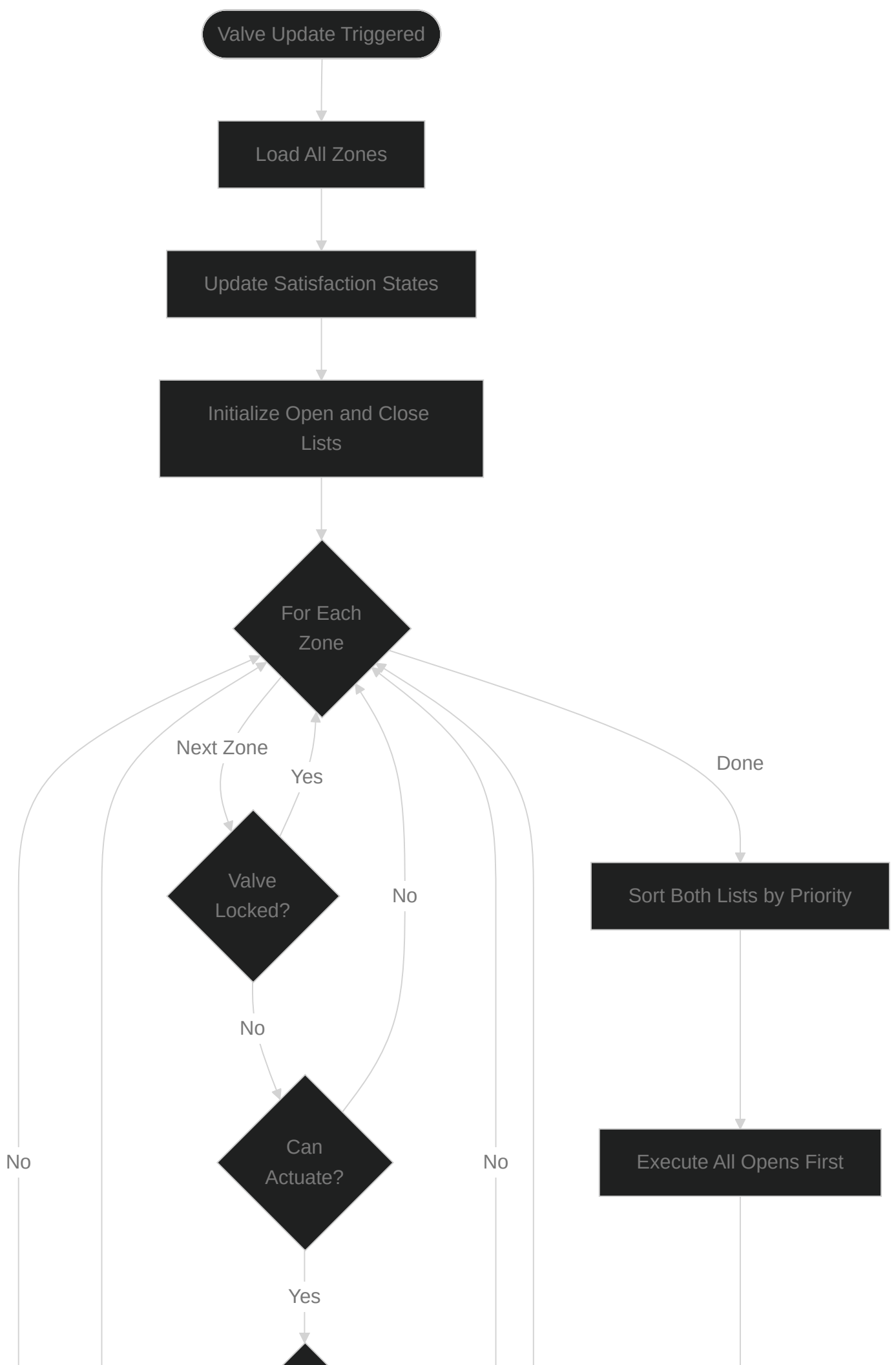
Yes

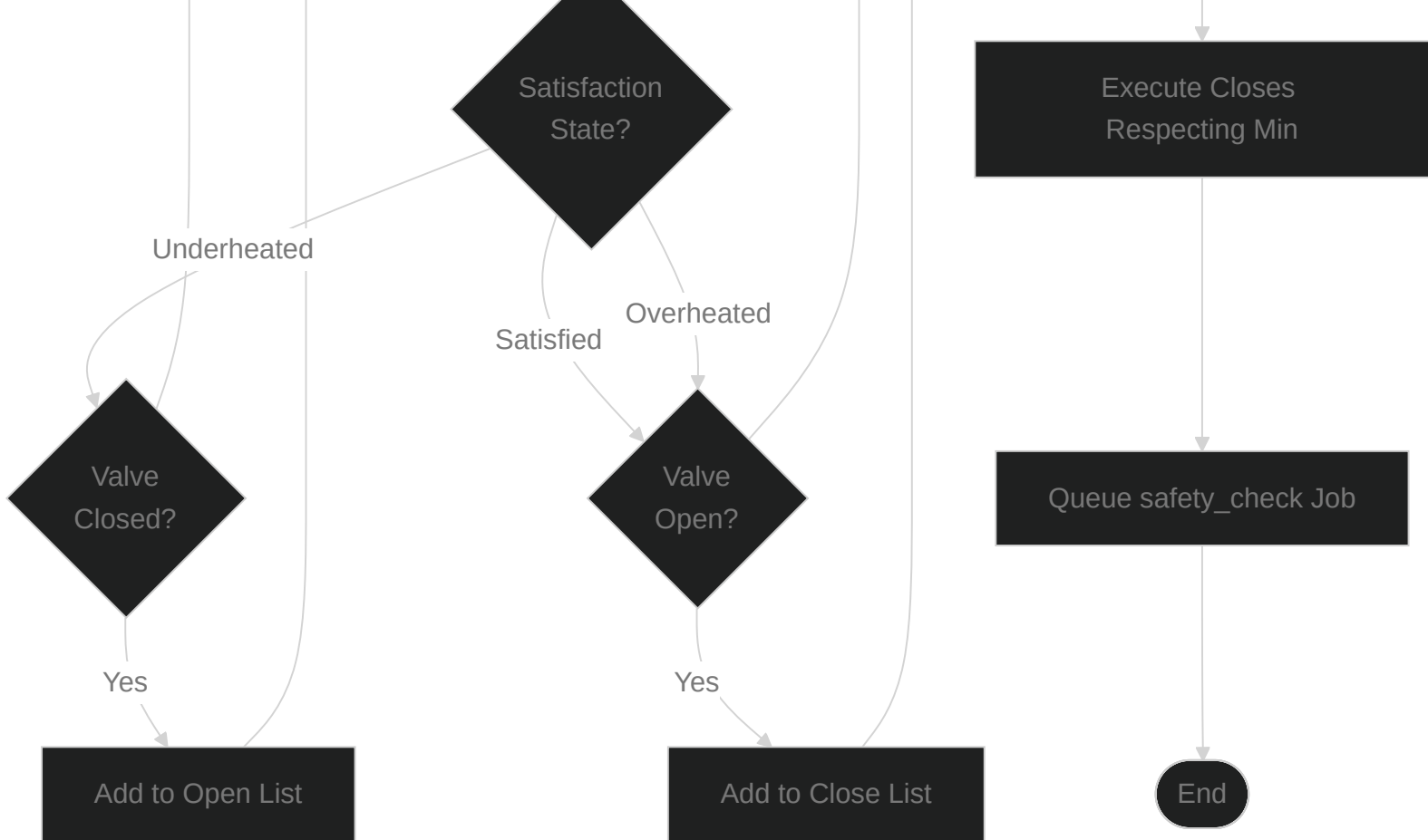
Store New Target in Redis

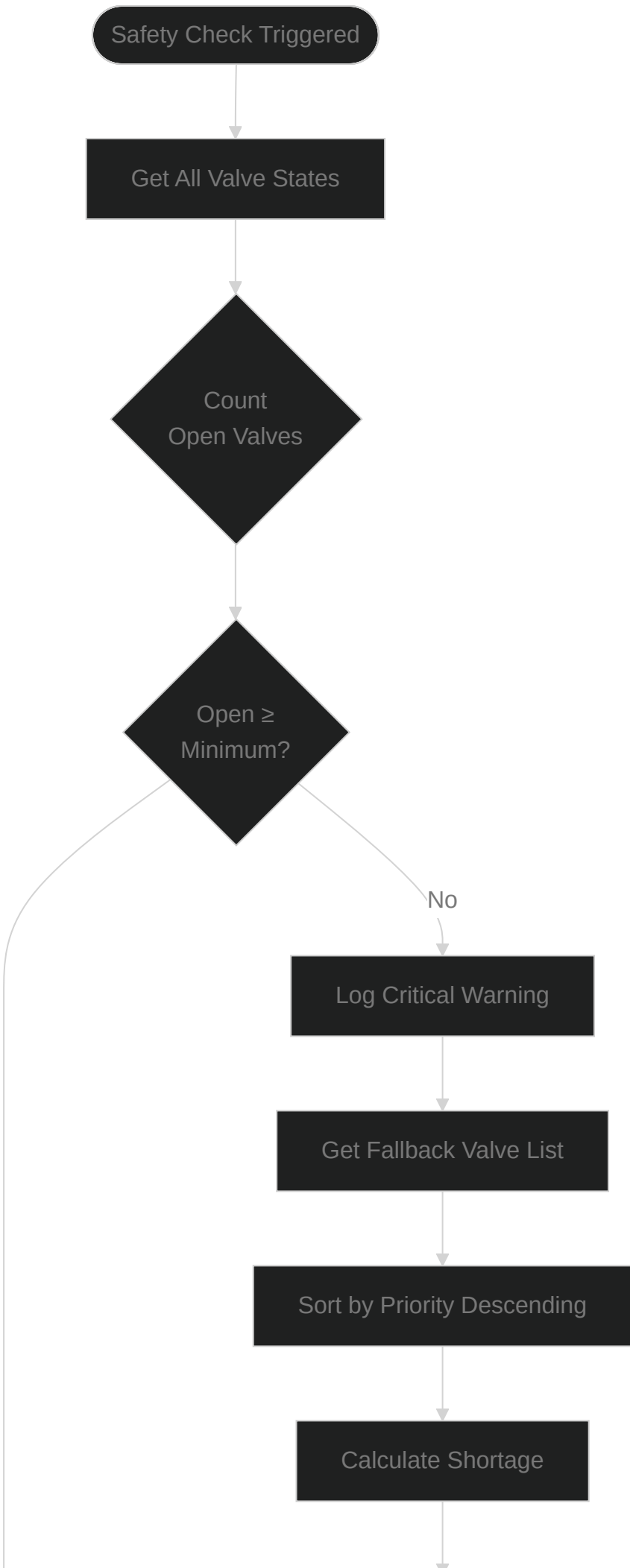
Queue Set Temp Command

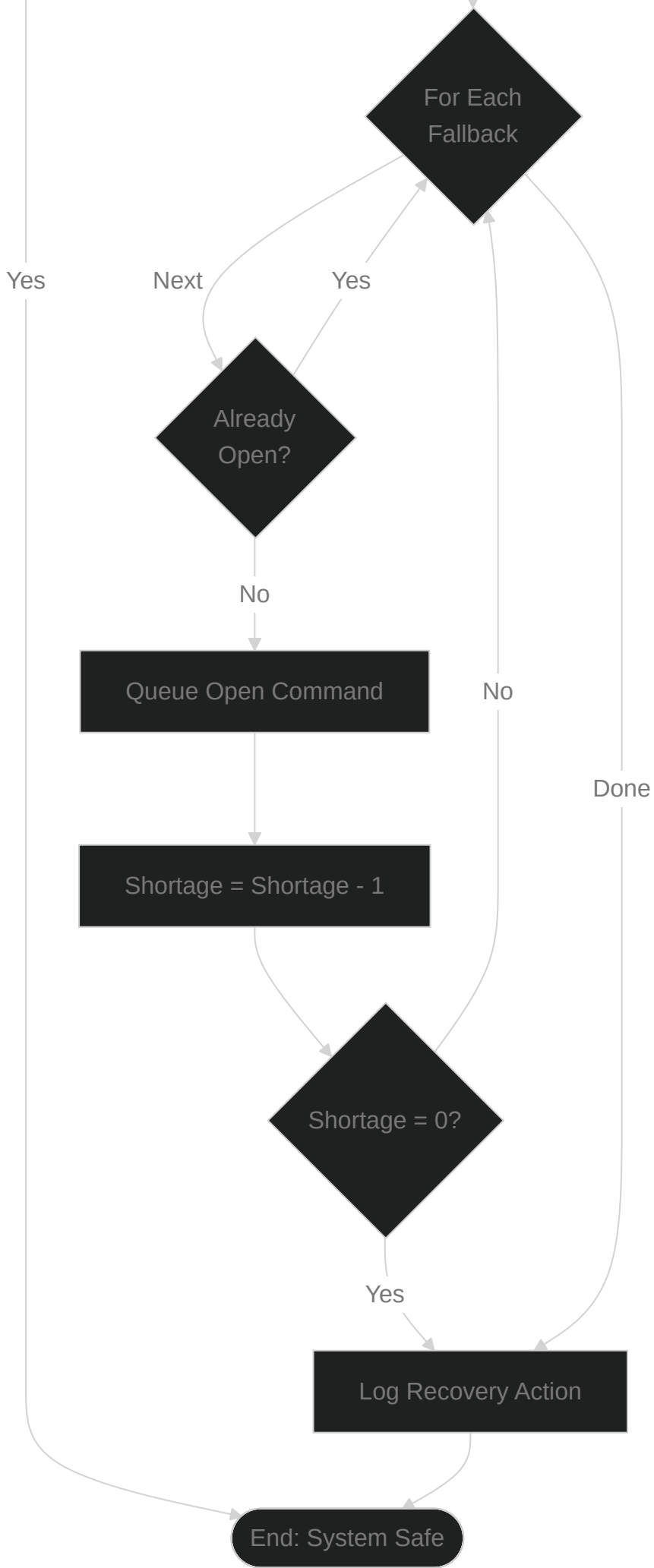
Queue update_valves Job

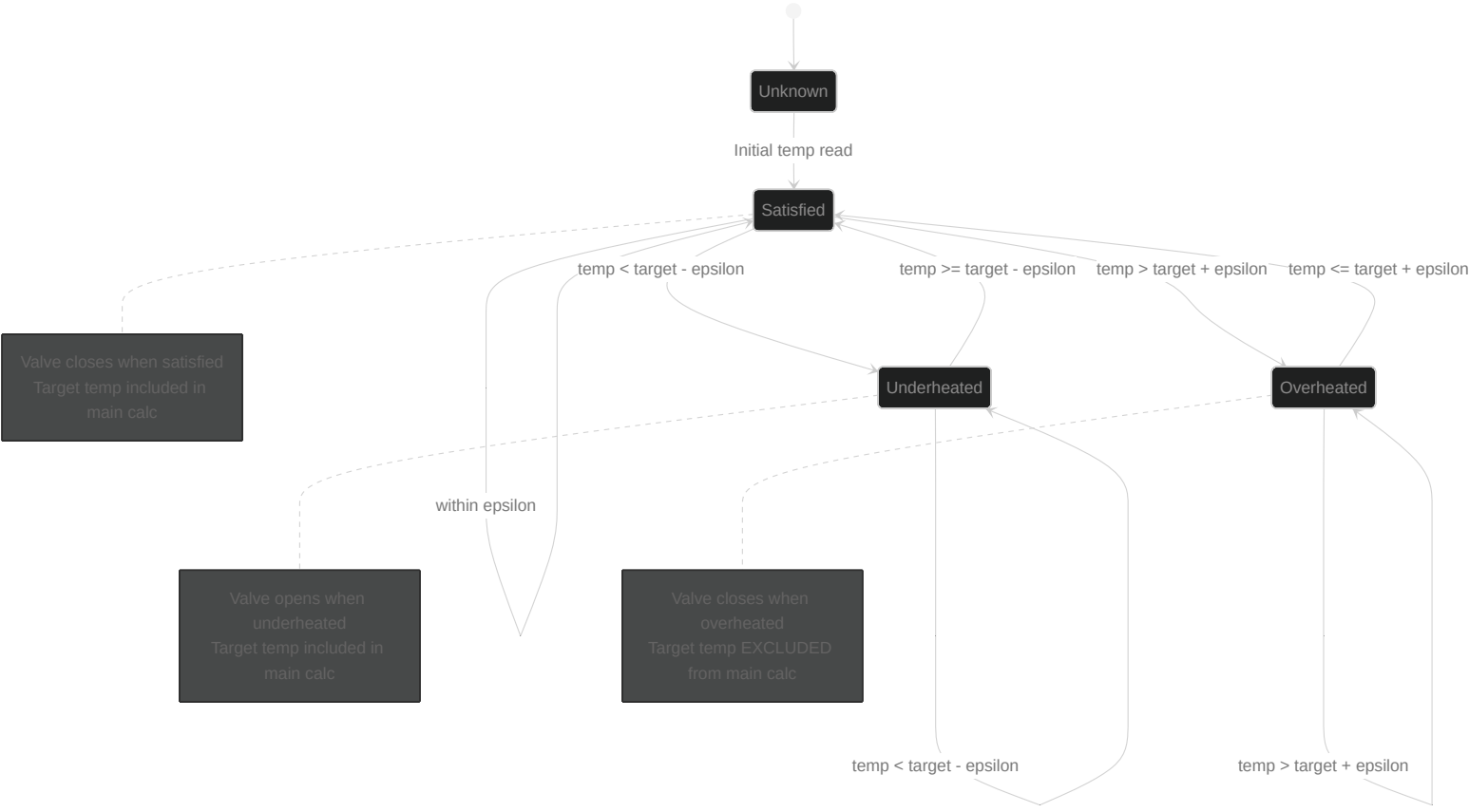
End: No Update

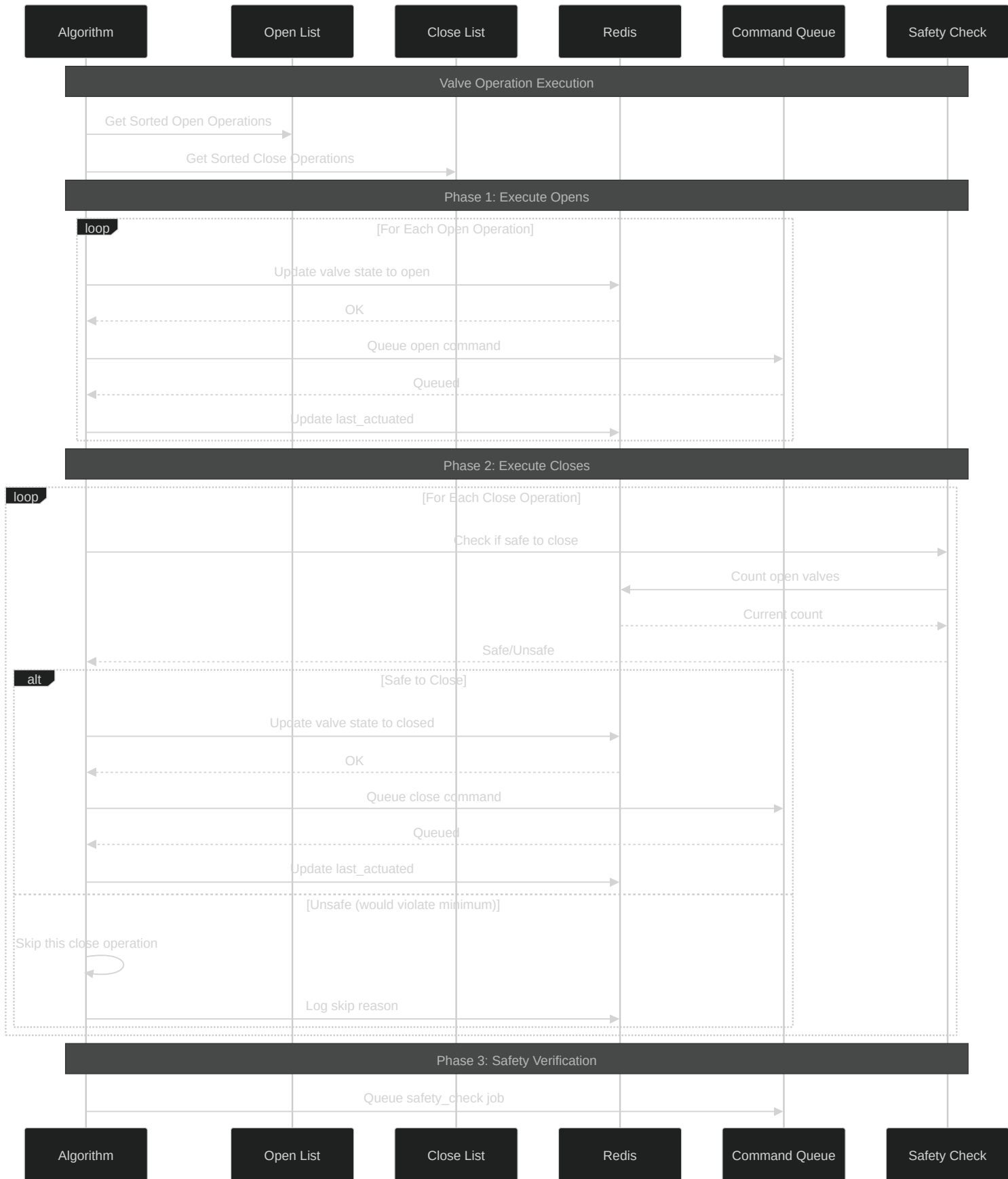












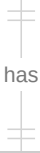
GLOBAL_CONFIG	
bool	use_average_mode
float	slider_position
float	min_temp
float	max_temp
float	change_threshold
int	min_valves_open
int	valve_actuation_delay
float	satisfaction_epsilon

COMMAND		
string	id	PK
string	type	
string	target_entity	
json	parameters	
string	status	
timestamp	created_at	
timestamp	completed_at	

JOB_QUEUE	
string	job_type
json	payload
timestamp	queued_at



ZONE_CONFIG		
string	id	PK
string	name	
string	temp_sensor_entity	
string	valve_entity	
float	target_temp	
int	priority	
bool	is_fallback_valve	
bool	enabled	



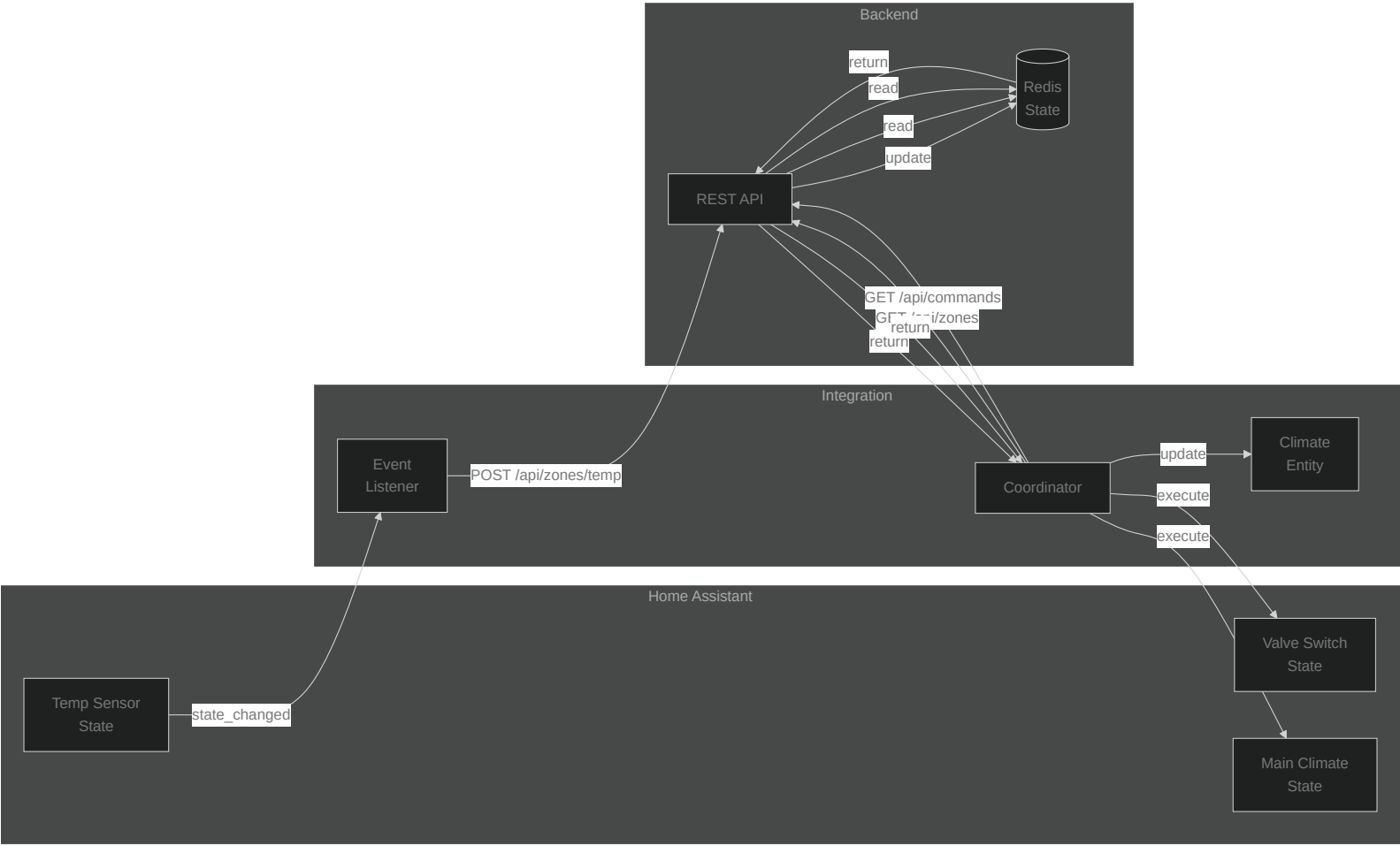
ZONE_STATE		
string	id	PK
float	current_temp	
string	satisfaction	
bool	valve_open	
timestamp	last_actuated	
timestamp	valve_lock_expiration	
timestamp	last_updated	

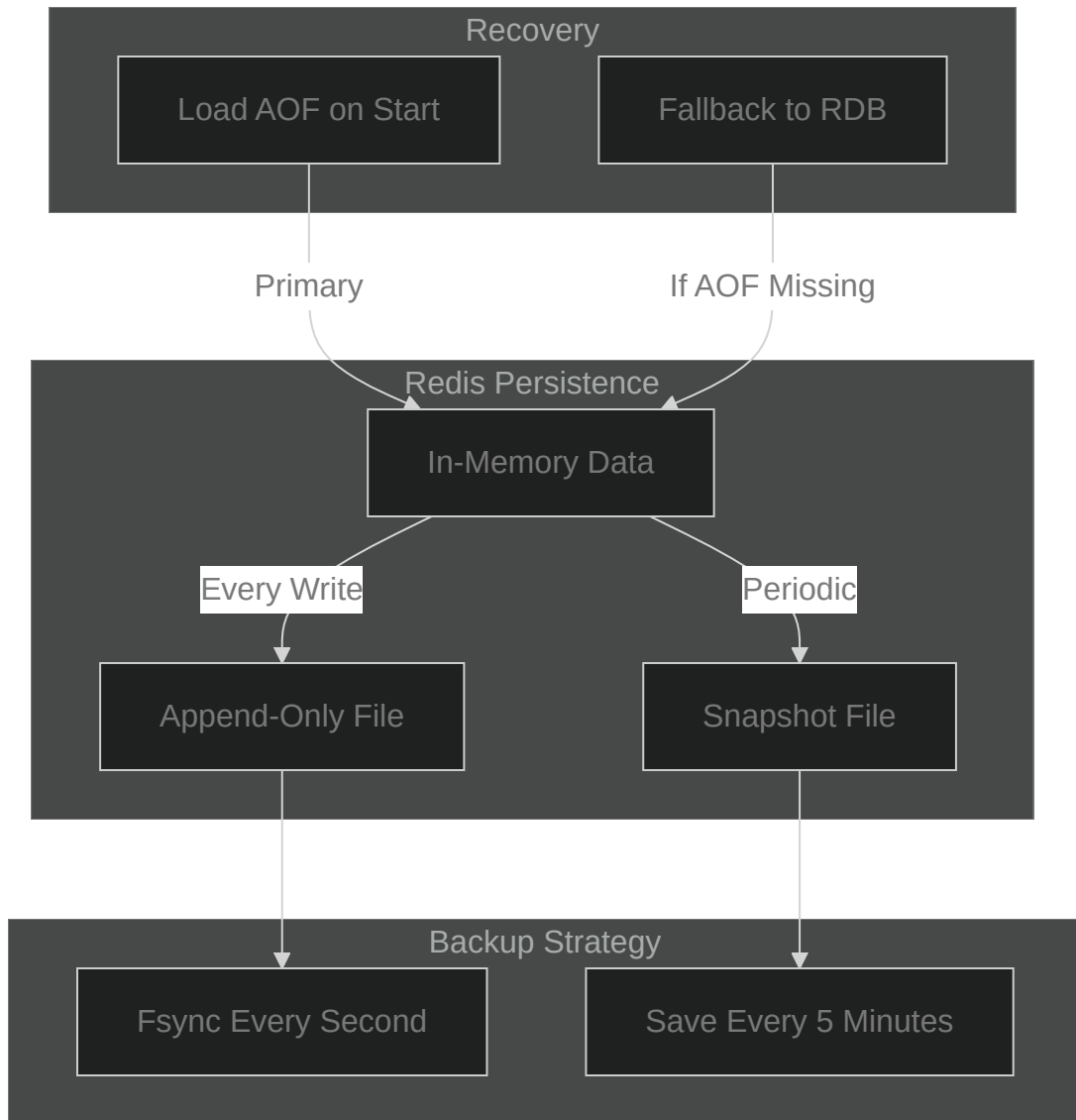


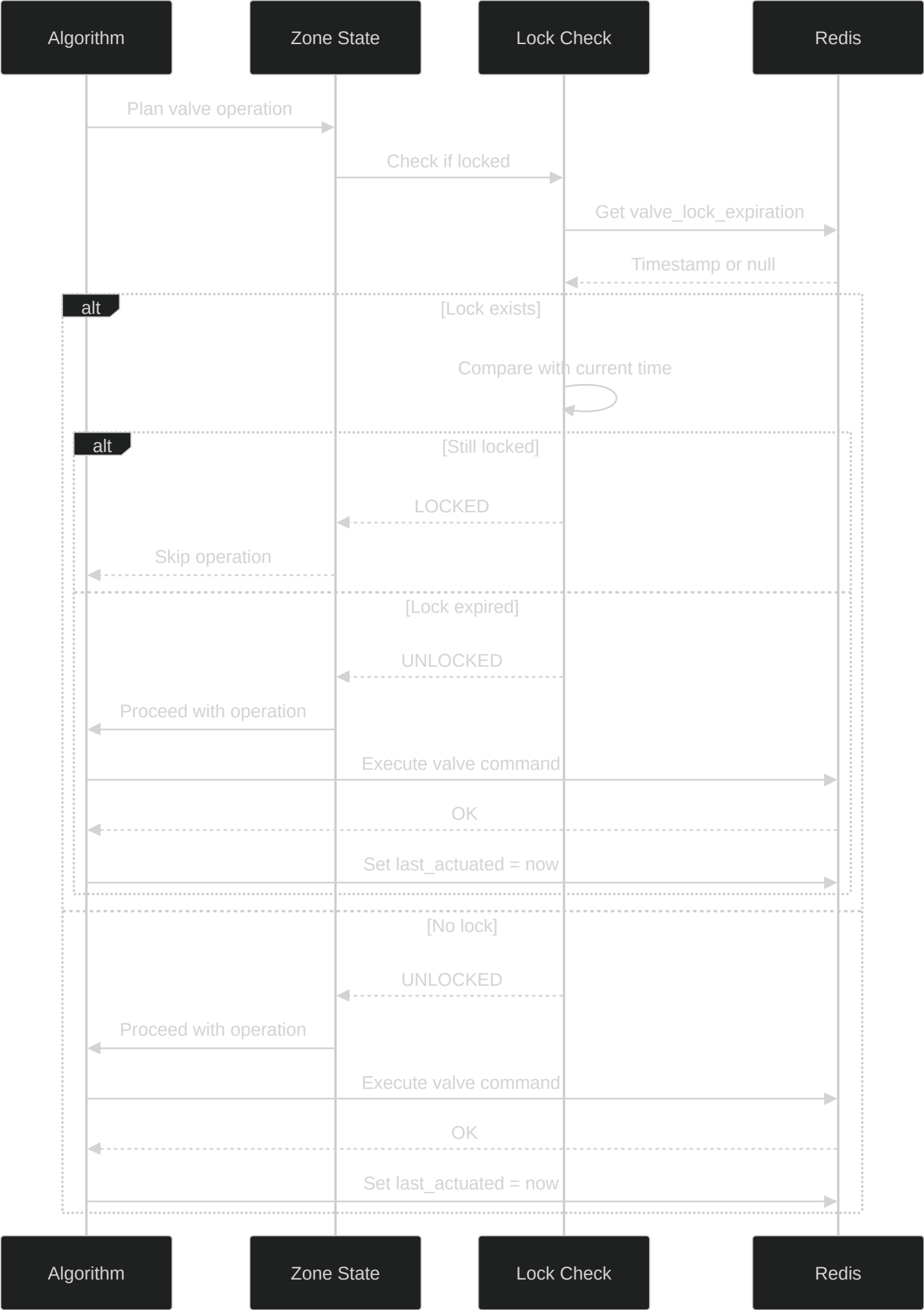
generates

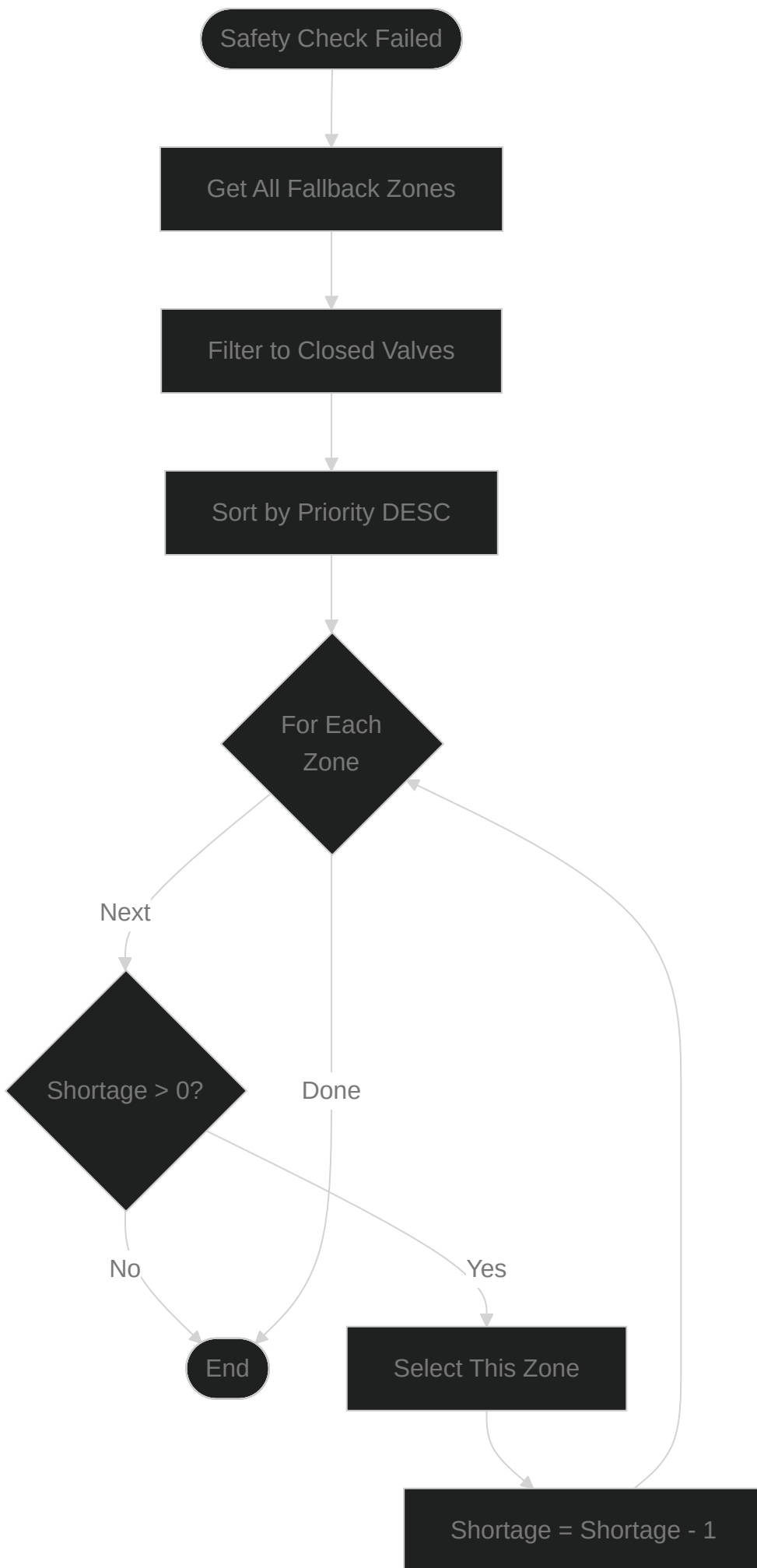


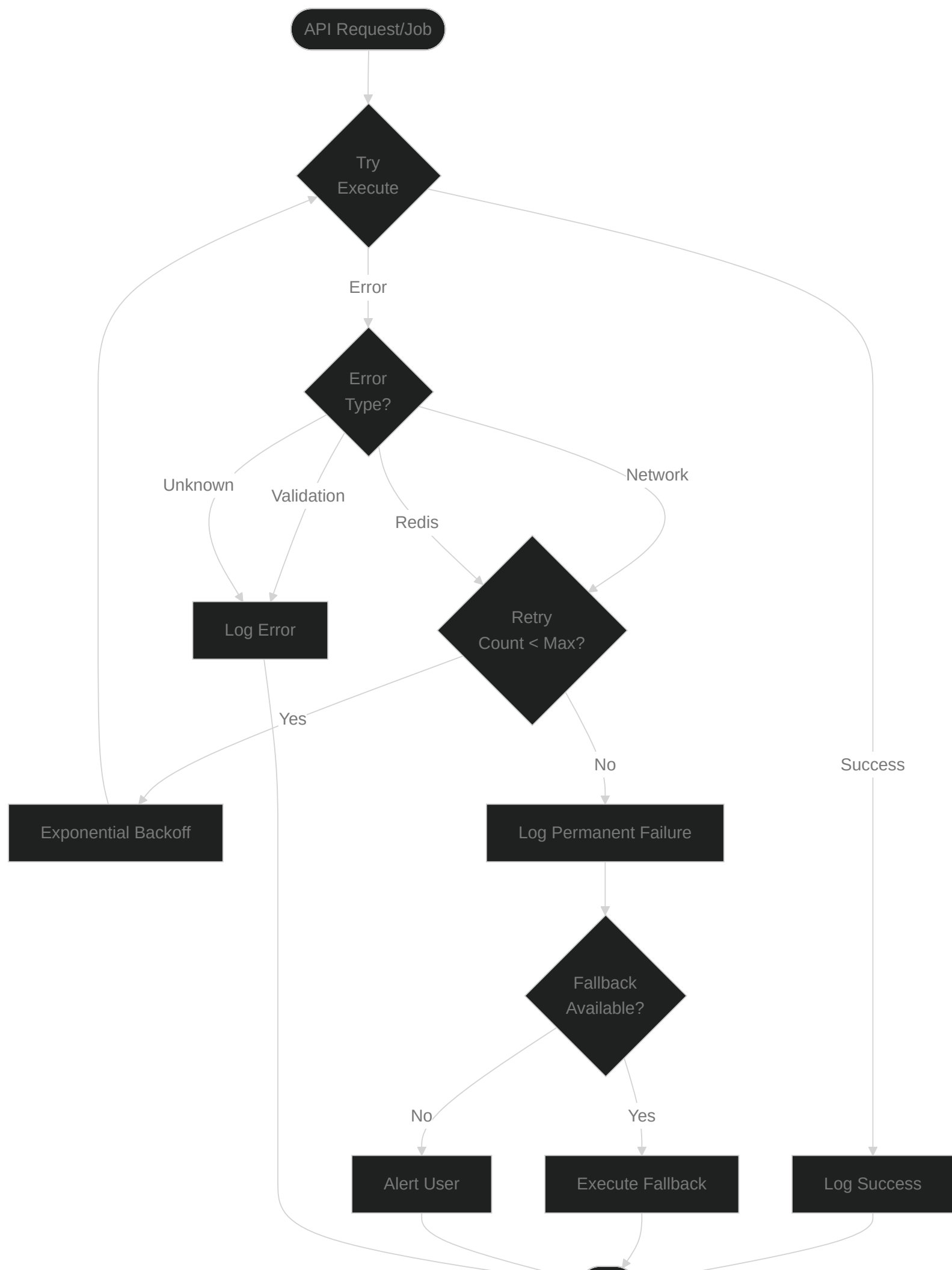
METRICS	
string	zone_id
timestamp	time
float	temp
string	state
bool	valve_state





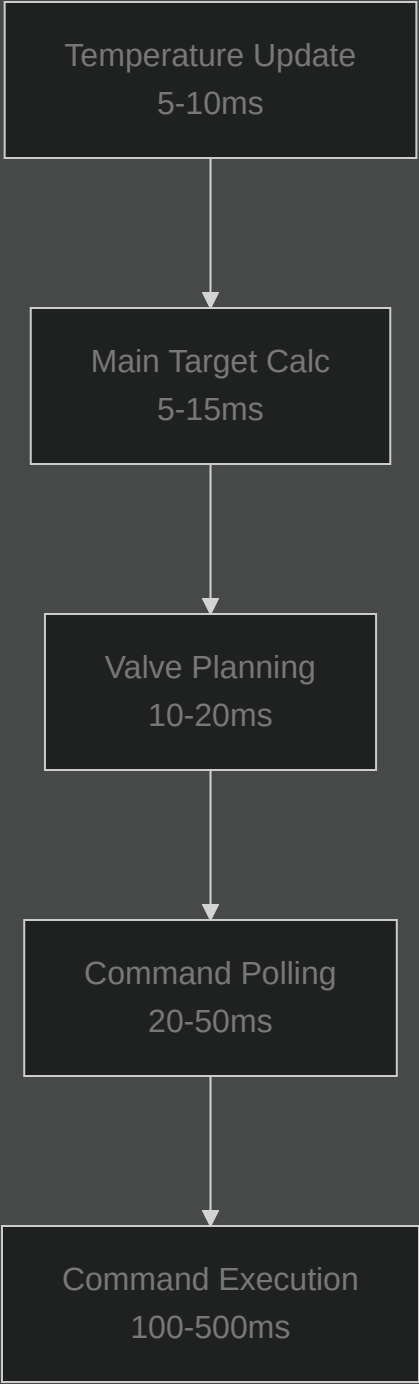


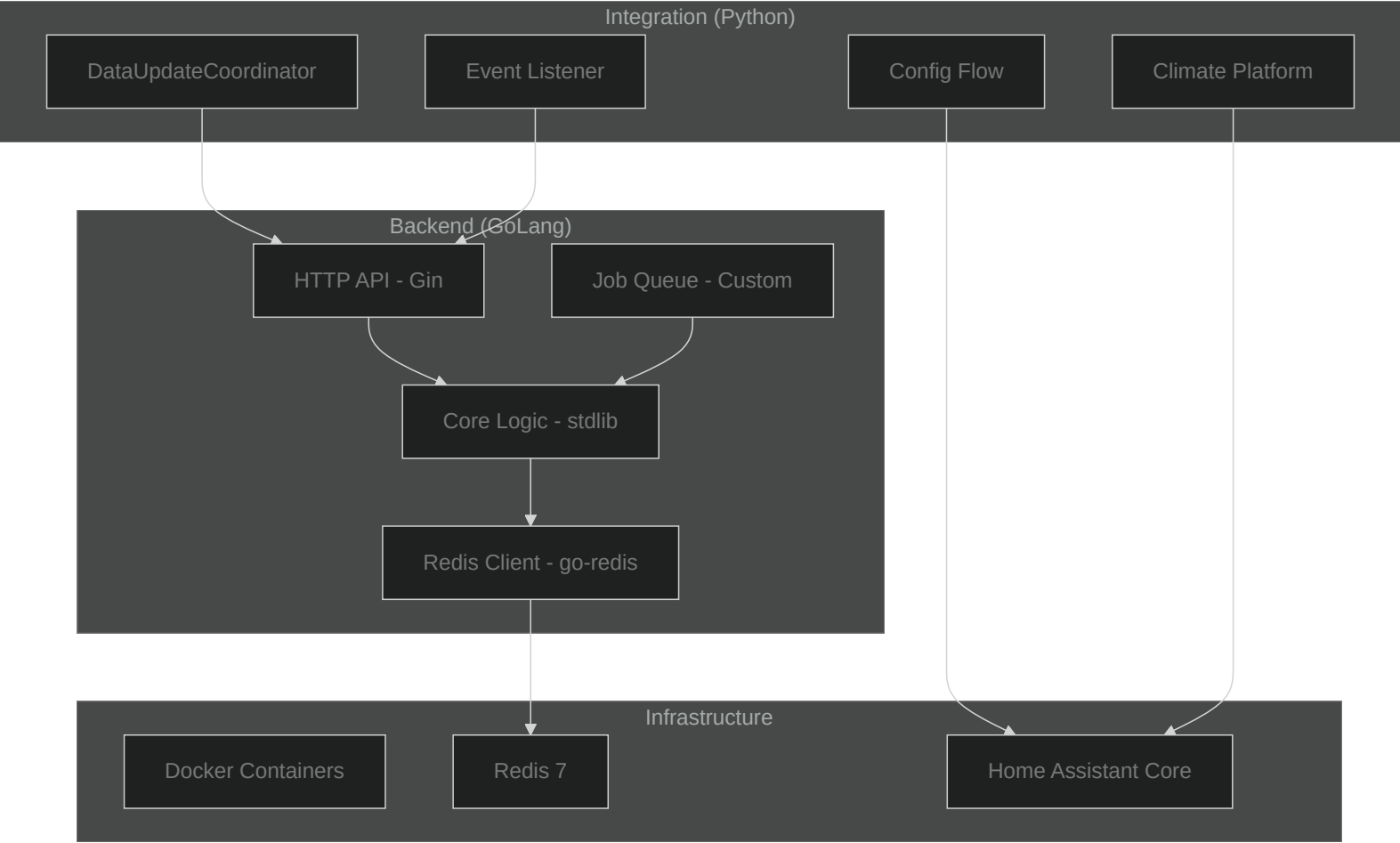






Latency Profile





Backend Tests

Unit Tests
Algorithm Logic

Integration Tests
Redis + API

Mock Tests
HA Service Calls

Integration Tests

Config Flow Tests

Entity Tests

Coordinator Tests

E2E Tests

Scenario Tests
Full Flow

Safety Tests
Edge Cases